ModelIt!

Teacher's Guide

How Do Seasons Change?

The Science of Seasonal Patterns

Quick Reference

Grade Level	Grade 1
NGSS Standard	1-ESS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 1-ESS1-2

Make observations at different times of year to relate the amount of daylight to the time of year.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students observe and analyze how daylight and temperature change across the four seasons.
Disciplinary Core Idea	**ESS1.B Earth and the Solar System** — Seasonal patterns of sunrise and sunset can be observed, described, and predicted.
Crosscutting Concept	**Patterns** — Students recognize that seasons follow a predictable, repeating pattern throughout the year.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The four seasons are one of nature's most important patterns. Spring, summer, fall, and winter bring changes in weather, daylight, and the natural world around us.

What Causes Seasons

Seasons happen because the Earth is tilted as it moves around the sun. During summer, your part of the Earth tilts toward the sun, so the sun is higher in the sky and shines for more hours each day. During winter, your part tilts away from the sun, so the sun is lower in the sky and shines for fewer hours. More hours of sunlight means more warmth, which is why summer is hot and winter is cold.

The Seasonal Pattern

Seasons always follow the same order: spring, summer, fall, winter. This pattern repeats every year without fail. In spring, days get longer and temperatures warm up. In summer, days are the longest and temperatures are the warmest. In fall, days get shorter and temperatures cool down. In winter, days are the shortest and temperatures are the coldest. Then spring comes again and the cycle starts over. People, plants, and animals all depend on this reliable pattern.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System

Students identify amount of daylight as the external component (it changes with the seasons and we cannot control it) and temperature as the internal response.

E — Establish Relationships

Students discover that when daylight increases, temperature increases — a positive relationship.

V — Visualize & Model

Students build a simple model connecting amount of daylight to temperature.

E — Evaluate Outcomes

Students test their model by comparing summer (high daylight) and winter (low daylight) scenarios.

R — Revise & Extend

Students reflect on how their model matches their own experience of warm summers and cold winters.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"What season is it right now? How do you know? What is different about summer, fall, winter, and spring?"*

Watch For

Show the four-season tree image. Students call out what they notice about each season.

Slide 2: What Will We Learn?

3 min

Say: *"Today we will learn WHY the seasons change. We will discover the connection between sunlight and temperature."*

Watch For

Introduce vocabulary: season and daylight. Ask: When is it light outside the longest — summer or winter?

Slide 3: Let's Be Scientists!

3 min

Say: *"Why is it hot in summer and cold in winter? What do you think causes the difference?"*

Watch For

Students share their ideas. Record predictions on chart paper to revisit at the end.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts: Amount of Daylight is how long the sun shines. Temperature is how hot or cold it gets."*

Watch For

Show a long yellow strip (summer daylight) and a short yellow strip (winter daylight). Ask: Which is warmer?

Slide 5: How Are They Connected?

5 min

Say: *"When there is MORE daylight, the temperature goes UP because the sun warms the Earth longer. More sun means more warmth!"*

Watch For

Students use sentence frame: When daylight goes up, temperature goes up. Draw the positive arrow together.

Slide 6: What Did We Find Out?

5 min

Say: *"We discovered that the amount of sunlight controls the temperature. Summer has the most sun and is warmest. Winter has the least sun and is coldest."*

Watch For

Revisit predictions. Show a graph of daylight hours across the year and trace the pattern together.

Slide 7: STEM Challenge

10 min

Say: *"Now build a season wheel that shows how daylight and temperature change! It should spin to show the pattern repeating."*

Watch For

Distribute paper plates, brass fasteners, and coloring supplies. Students create and present their season wheels.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Amount of Daylight
- **INTERNAL:** Temperature

Relationships

- **Amount of Daylight to Temperature:** POSITIVE (+) — More hours of sunlight warm the Earth more, raising the temperature. In summer, long days make it hot. In winter, short days mean less warming and colder temperatures.

Key Discoveries

- Seasons follow a pattern: spring, summer, fall, winter, and then it starts again
- Summer has the most daylight hours and the warmest temperatures
- Winter has the fewest daylight hours and the coldest temperatures
- More daylight means warmer weather because the sun warms the Earth longer

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Seasons happen because the Earth is closer to the sun in summer"	The Earth's distance from the sun does NOT cause seasons. In fact, Earth is actually slightly closer to the sun during January in the Northern Hemisphere! Seasons are caused by the Earth's tilt, which changes how directly sunlight hits your part of the Earth and how many hours it shines.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"It is cold in winter because the sun is turned off or weaker"	The sun shines just as strong in winter as in summer! The difference is that in winter, your part of the Earth is tilted away from the sun. The sun is lower in the sky and out for fewer hours, so it has less time to warm the ground. The sun is always powerful — it just reaches us differently in winter.